

Exercise series 6: Active Databases



ADReM
Advanced Database Research & Modelling
University of Antwerp

 Universiteit
Antwerpen

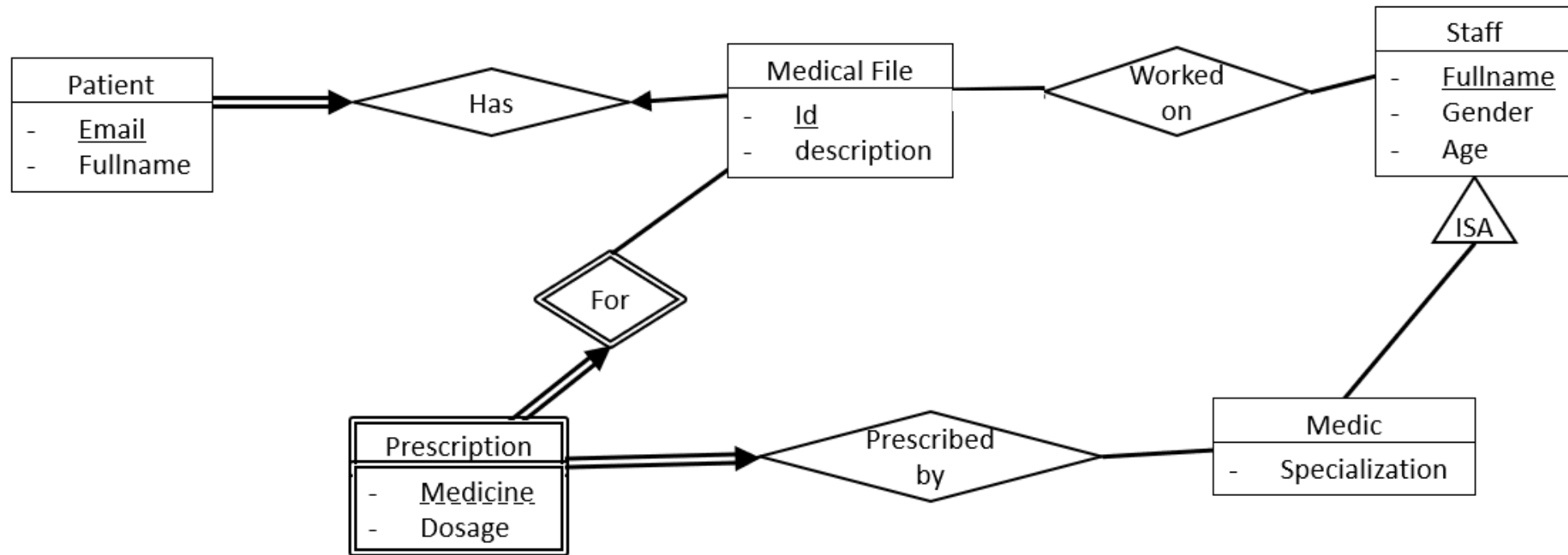
Problem 1

Create in PostgreSQL a schema based on the following description.

- A student has a first and last name. The combination of first and last name must be unique, and both fields are mandatory. A student can also be identified by a unique enrolment number which must be between 10^*000 and 99^*999 .
- A course has a unique name and id, which must be a positive integer. A course also has a number of student points. The number of student points is either 3, 6, 9 or 12.
- An academic year consist of a starting year and ending year. The starting year should be a year before the ending one, and both should be between 2000 and 2100.
- A professor is identified by the combination of first and last name.
- A course offering is composed of a course, an academic year and a professor. We also store the semester: either 'first' or 'second'. The combination of year, semester and course identifies the course offering.
- If student enrolment number, professor name or course id is updated, all references are updated.
- If a student is deleted, delete all enrolments of that student.
- If a professor is deleted set reference to null.
- If a course is deleted, delete all enrolments and teaching assignments.

Problem 2

Consider the following ER diagram for a private clinic:



Problem 2

1. Translate the ER diagram into a relational model

2. Use checks, constraints and domains to enforce the followings constraints:

- Dosage is in *mg* and should be an integer greater than zero.
- A patient can be identified by his/her name
- Staff age must be above or equal 18 y.o.
- If a medic is removed then his/her prescriptions are not valid anymore and must be removed. If a medic changes name then it is updated.
- If a staff member is removed or changes name then all the entries with a references to that member are deleted/updated also
- If a patient removes his/her account then his/her medical file is kept but without the reference.

